

Process Mining and Formal Verification of the Volvo IT Incident Process

BPI Challenge 2013

Formal Methods

September 7, 2026

- An IT service desk records every state change of every incident.
- The process is *documented*. What was actually *executed*?
- Volvo IT Belgium, VINST system, 2010–2012.
- Three logs: incidents, open problems, closed problems.
- A standard benchmark, so the results are comparable with the literature.

Log	Cases	Events	Activities	From	To
closed_problems	1,487	6,660	7	2006-01-11	2012-05-31
incidents	7,554	65,533	13	2010-03-31	2012-05-22
open_problems	819	2,351	5	2006-11-07	2012-06-15

The problem logs are not variants of the incident process — median duration differs by an order of magnitude.

There is no activity column

The dataset records Status and Sub Status. What counts as an activity is the analyst's decision.

Status — 4 activities

Accepted, Queued, Completed, Unmatched

Small models. Waiting on a vendor and waiting on a user become indistinguishable.

Status+Sub Status — 13 activities

Accepted+In Progress, Accepted+Wait - User, ...

Three times larger models, lower precision, and the real process.

Treated as a parameter, not fixed silently. Both are reported.

Three discovery algorithms

- Alpha** Ordering relations from directly-follows pairs. Assumes a complete, noise-free log. Cannot represent short loops.
- Heuristics** Frequency-weighted, with a dependency threshold. Filters infrequent behaviour instead of modelling it.
- Inductive** Recursive partitioning into a process tree. Guarantees soundness and, at zero noise, perfect fitness.

Four dimensions, not one

Fitness Can the model replay what happened?

Precision Does it forbid what never happened?

Generalisation Will it hold for future cases?

Simplicity Can a human read it?

They trade off. Maximising any one alone is trivial — a model permitting everything fits perfectly and explains nothing.

Mode	Algorithm	Fitness	Precision	Gen.	Simpl.	Places	Trans.
status	alpha	0.119	0.400	0.734	1.000	2	4
status	heuristics	0.994	0.678	0.767	0.622	8	15
status	inductive	1.000	0.440	0.748	0.667	17	23
status+substatus	alpha	0.127	0.176	0.770	0.789	2	13
status+substatus	heuristics	0.948	0.387	0.634	0.471	27	69
status+substatus	inductive	1.000	0.234	0.783	0.634	57	78

Inductive buys perfect fitness with permissiveness and the largest net. Heuristics gives the best balance. Alpha collapses.

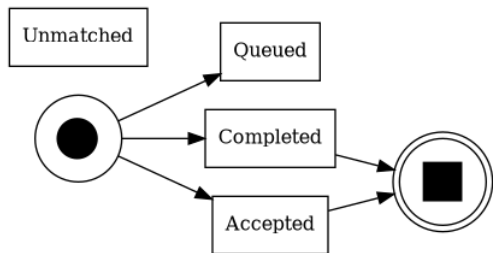
Why Alpha collapses

Fitness below 0.15; the net degenerates to two places. **This is not a bug.** 90% of cases contain a

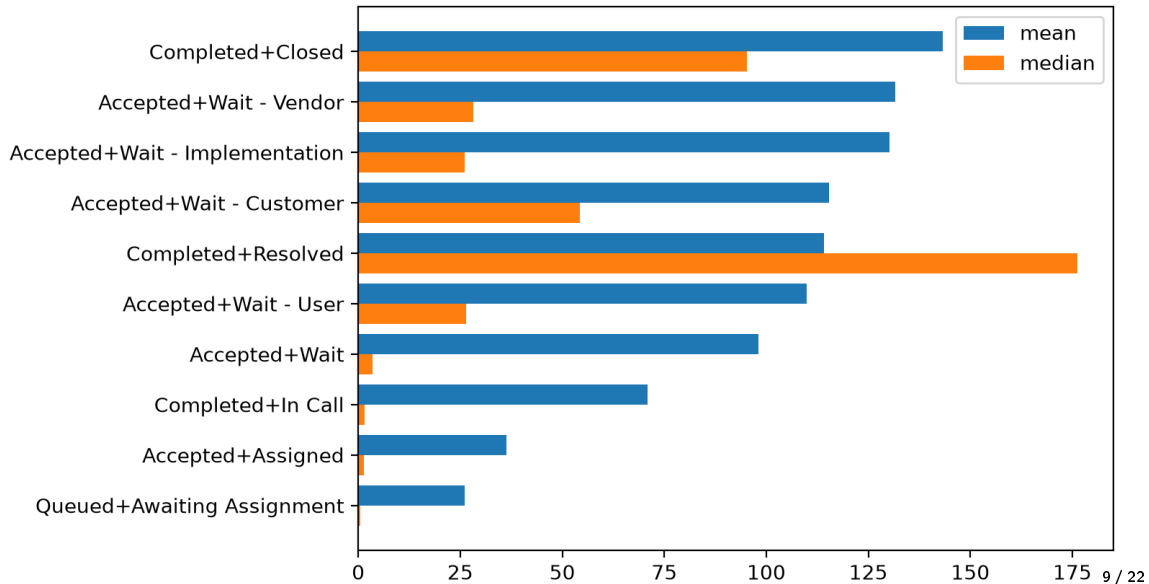
length-one loop — an activity immediately repeated. Alpha infers contradictory ordering relations from a

directly-follows pair (a, a) and provably cannot represent such loops. A known theoretical boundary,

demonstrated on a real log.



Where does the time go?

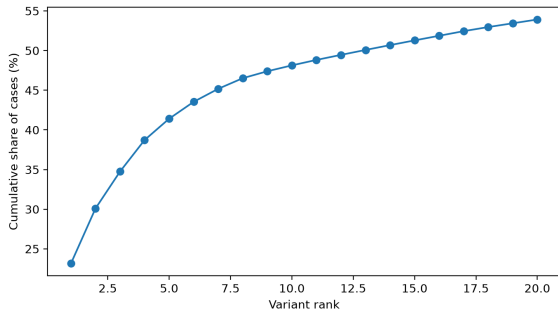


Accepted+In Progress: median waiting time in **minutes**, mean in **hours**.

- Work is normally picked up immediately.
- Occasionally it stalls for a very long time.
- The mean alone says “slow activity”. The median alone hides the stalls.

Both are reported everywhere in this analysis.

A weakly standardised process



Thousands of distinct variants against 7,554 cases. Most variants occur exactly

once. Consequence: any high-precision

model of this log would be overfitted. The low precision figures are the correct result.

A case is a finite sequence of events, so the semantics are LTL_f .

$$\pi, i \models F\varphi \iff \exists j \geq i. \pi, j \models \varphi$$

$$\pi, i \models G\varphi \iff \forall j \geq i. \pi, j \models \varphi$$

$$\pi, i \models X\varphi \iff i + 1 < n \text{ and } \pi, i+1 \models \varphi$$

X is *strong*: false at the last position. Remember this for two slides' time.

Eight properties, each a real formula evaluated per case.

Termination	$F(\text{completed}_*)$
Resolution	$G(\text{in_progress} \rightarrow F \text{ completed}_*)$
No work after closure	$G(\text{closed} \rightarrow \neg F \text{ in_progress})$
Queued work picked up	$G(\text{queued} \rightarrow F \text{ in_progress})$
Formal closure	$G(\text{in_progress} \rightarrow F \text{ closed})$
Queueing precedes work	$(\neg \text{in_progress}) \cup \text{queued}$

The last two encode the *documented* procedure. They are expected to fail. The size of the gap is the result.

Property	Satisfied	Violated	Ratio
Termination	7,546	8	0.999
Resolution	7,545	9	0.999
No work after closure	7,435	119	0.984
Queued work is picked up	7,494	60	0.992
Waiting on the user resumes	7,550	4	0.999
No unmatched events	7,549	5	0.999
Formal closure	5,591	1,963	0.740
Queueing precedes work	1,168	6,386	0.155

Fewer than half of all events are the *first* occurrence of their activity within the case.

- Over half of all recorded work is repetition.
- Completion is near perfect; variability is moderate.
- Rework is the single component dragging the score down — and the largest available lever.

$(\neg in_progress) \cup queued$ holds for a small minority of cases.

- The documented procedure: queue the incident, then work it.
- What happens: work begins without the incident ever being queued.
- Established by *formal verification over every case* — not by sampling, not by eyeballing variants.

Verification

$G(\text{closed} \rightarrow \neg F \text{ in_progress})$ fails for a set of cases: work resumes after formal closure.

Performance

Completed+Closed appears in the waiting-time table at all — meaning closure events that were *not* the last event of their case.

Independent methods agreeing is stronger than either alone.

The strong-X trap

Written the natural infinite-trace way:

$$G(\text{closed} \rightarrow X G \neg \text{in_progress}) \implies 26.2\%$$

Written without the next operator:

$$G(\text{closed} \rightarrow \neg F \text{in_progress}) \implies 98.4\%$$

Most cases *end* with Completed+Closed. X is false there, so every such case falsifies the first formula for reasons having nothing to do with Volvo IT.

A factor of four apart. Only one of them is about the process.

Four stateless layers

Loading every source attribute retained

Mining discovery, conformance

Verification LTL_f per trace

Analysis time, variants, anomalies

Consequences

- Testable without a web server
- Regression tests pin every figure shown today
- Interactive dashboard over the same functions
- Every table and figure generated from code

- Conformance is computed on a *sample*; alignment-based computation does not scale to the full log.
- Token-based replay is optimistic on unsound nets.
- The composite score's weights are *stipulated*. Components are meaningful; the aggregate only comparatively.
- Unmatched+Unmatched is an export artefact, not work.
- We observe what was *recorded*. Poor recording discipline looks like rework.

- ① **Rework dominates** — over half of recorded work is repetition.
- ② **Documented $\neq$ executed** — push-to-front proved formally over every case.
- ③ **Alpha is inapplicable here** — predicted by length-one loops, not an empirical surprise.

Next: the organisational attributes are already loaded — handover analysis across support lines, and incident versus problem as interacting processes.

Questions